

Somaiya Vidyavihar University
K. J. Somaiya School of Engineering

Evaluation scheme for laboratory/tutorial continuous assessment (LAB CA)

Branch	EXTC	Year	SY
Academic year:	Jan 2026 - April 2026	Course Code and Name	216U03C401 Probability, Statistics and Optimization Techniques
Semester:	IV		
Faculty In-charge	Mrs. Sharvari Deshmane		

Distribution of LAB CA marks	The student will be evaluated based on following tasks for lab. CA. If any of the tasks given is not completed / submitted / shown / evaluated, then the marks assigned for that task will be ZERO.			
	Task	Description	Tentative schedule (week/month)	Marks
	Laboratory experiment	Experiments will be performed in the laboratory	Each week of semester	25
	Test based on laboratory experiments	Test will be conducted Batch wise during the regular lab	Last week of March 2026	10
	Mini Project (Group Activity)	• Concept Selection: • Computational Modeling: Write Python code using NumPy, SciPy, or Pandas to simulate thousands of random trials/events. • Visualize the results • Analysis: Compare the Simulated Result against the Theoretical Value	Second/Third week of April 2026	15

Calculation of final LAB CA marks (out of 50):

Component 1	Component 2	Component 3
Laboratory experiments	Test	Mini Project
Maximum marks: 25	Maximum marks: 10	Maximum marks:15
Final LAB CA Marks (Out of 50) = Component 1 (out of 25) + Component 2 (out of 10)+ Component 3 (out of 15)		

Date: 21/01/2026

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Evaluation Rubrics for Laboratory experiments:

Criterion/ Performance levels	Performance level 1	Performance level 2	Performance level 3
Laboratory ethics and organization of laboratory work (10 Marks)	Active in Laboratory. All work submitted on time, no delays. Exceptionally clear, detailed, and well-organized write-ups. Proper Data Visualization (7-10)	Partially active in Laboratory. Work submitted late with valid reasons. Generally clear write-ups but lacking in detail or organization of Data Visualization (2-6)	Passive in Laboratory. Late submissions, without valid reasons. Write-ups are unclear, poorly organized, and lack Data Visualization (0-1)
Performance of laboratory experiments (10 Marks)	Correct use of statistical functions. Formulas are accurately translated to code. (7-10)	Conducted experiments with some errors but generally correct. Basic understanding of theory, several gaps in knowledge. (3-6)	Frequently made errors, affecting the experiment outcomes. Poor understanding of relevant theory. (0-2)
Conclusion, Ability to answer questions related to the respective practical (05 Marks)	Can explain the "Statistical Significance" of the Colab output. Strong link between code and theory. (4-5)	Understands the output but needs prompting to link it back to theoretical probability. (2-3)	Can explain what the code does, but cannot explain what the results mean statistically. (0-1)

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Evaluation Rubrics for Mini Project (Using Python): Group of 3-4 students are required to simulate small concept/project and demonstrate the same to laboratory teacher.

Criteria	Excellent (5 Marks)	Good (4-3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1-0 Marks)
Concept & Simulation Logic	Flawless simulation of the statistical concept; parameters are correctly defined and used.	Concept is mostly correct; minor logic errors that don't affect the overall outcome.	Basic simulation; logic is oversimplified or has noticeable gaps.	Simulation logic is incorrect or does not reflect the chosen concept.
Python Implementation, Data Visualization & Results	Efficient, clean code. No errors. Excellent use of Matplotlib / Seaborn. Clear labels, titles, and insightful graphs.	Code runs well; but could be more optimized. Graphs are present and correct but lack detailed labelling or clarity.	Code runs but is poorly structured; relies on hard-coded values. Basic plots are used but are difficult to interpret or visually messy.	Code has bugs or fails to execute during the demonstration. No visualization provided, or plots are irrelevant to the concept.
Viva & Team Coordination	Both members explain the logic clearly. Deep understanding of the math-code link.	Good explanation by both, but one member is significantly more dominant.	Can answer basic questions but struggles with the "why" behind the code.	Unable to explain the code or the statistical concept being simulated.

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